GENERAL NOTES

CONCRETE NOTES

1. REINFORCING BARS THAT ARE TO BE WELDED SHALL BE OF A WELDABLE GRADE AND WELDED IN ACCORDANCE WITH A.W.S. RECOMMENDATIONS.
2. PROTECTIVE COVERING OF REINFORCEMENT (SEE DETAILS) SHALL BE AS FOLLOWS:
   A. FOOTINGS AND GRADE BEAMS 3" CLEAR BOTTOM AND SIDES, 1 1/2" CLEAR SIDES.
   B. BEAMS 1 1/2" CLEAR TO STIRRUPS.
   C. CONCRETE COLUMNS AND PIERS 1 1/2" CLEAR TO TIES
   D. 3/4" CLEAR FOR CONCRETE JOIST TOP BARS
3. WHERE THE LENGTH OF A BAR IS GIVEN, AND IT IS TO BE HOOKED, THE HOOK SHALL BE IN ADDITION TO THE LENGTH GIVEN.
4. WHERE OPENINGS OCCUR IN SLABS, PLACE THE REINFORCING THAT WOULD OCCUR IN LINE WITH THE OPENING EQUALLY TO EITHER SIDE OF THE OPENING. CUT NO STEEL IN THE FIELD.
5. BACKFILL SHALL NOT BE PLACED AGAINST BASEMENT WALLS UNTIL THE BASEMENT FLOOR SLAB AND THAT PORTION OF THE FIRST SLAB OVER THE BASEMENT ARE IN PLACE.

CONTRACTOR'S NOTES

1. SEE ARCHITECTURAL, MECHANICAL, PLUMBING AND ELECTRICAL DRAWING. AND SPECIFICATIONS FOR SIZE AND LOCATION OF DRIPS, CHAMFERS, SLEEVES, ANCHORS, INSERTS AND OPENINGS REQUIRED. THE LOCATION AND SIZE OF SLEEVES OR OPENINGS NOT SHOWN ON THE DRAWINGS IN STRUCTURAL MEMBERS SHALL BE SUBJECT TO THE APPROVAL OF THE ARCHITECT. ANY CONFLICT SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT BEFORE PROCEEDING WITH THE WORK AFFECTED. PRINCIPAL OPENINGS IN THE STRUCTURE ARE INDICATED ON THE CONTRACT DRAWINGS. ANY SUBSTITUTIONS RESULTING IN REVISIONS TO THE STRUCTURE SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO COORDINATE WITH THE ARCHITECT.
2. THE CONTRACTOR SHALL NOT SCALE THE CONTRACT DRAWINGS FOR THE PURPOSE OF ESTABLISHING CORRECT DIMENSIONS.
3. U.N.O. DENOTES: "UNLESS NOTED OTHERWISE".
4. GENERAL CONTRACTOR SHALL FIELD VERIFY EXISTING SIZES, DIMENSIONS, NOTES OR CONDITIONS PRIOR TO ANY DETAILING, OR FABRICATION OF MATERIALS.
5. WHERE THE CONTRACTOR ELECTS TO POUR THE SLAB ON GRADE PRIOR TO ERECTING THE SUPERSTRUCTURE, THE CONTRACTOR SHALL TAKE INTO CONSIDERATION THAT THE SLAB ON GRADE IS NOT DESIGNED FOR SUPPORT OF ANY CRANES OR SIMILAR DEVICES USED TO ERECT THE STRUCTURAL FRAME. DO NOT PLACE THIS EQUIPMENT ON THE SLAB.
6. WHERE OVERHEAD LIFTS OCCUR ON THE PROJECT, NO KEYED OR SAW CUT JOINT SHALL BE PLACED WITHIN 6" OF THE LIFT SUPPORT ANCHORAGE AS PER THE LIFT MANUFACTURER RECOMMENDATIONS. GENERAL CONTRACTOR COORDINATE WITH FINAL LIFT PLACEMENT PLAN PRIOR TO POURING THE SLAB.
7. GENERAL CONTRACTOR SHALL ENGAGE A SURVEYOR TO PROVIDE LOCATIONS OF ALL EXISTING UTILITES, TRENCHES, ETC. TO ENSURE THAT NEW FOUNDATIONS WILL NOT INTERFERE, UNDERMINE, OR BEAR ON EXISTING UTILITIES.

DECKING NOTES

1. NO PIPING, DUCTWORK, OR CONDUIT LARGER THAN 3/4" DIAMETER OR STUD WALLS SHALL BE SUPPORTED DIRECTLY FROM METAL DECK OR PERMANENT METAL FORM.
2. ALL METAL ROOF DECKS SHALL BE WIDE RIB AND GALVANIZED TYPE "B" DECKS UNLESS NOTED OTHERWISE.

FOUNDATION NOTES

1. BUILDING FOOTINGS ARE DESIGNED TO BEAR ON ORIGINAL EARTH OR LABORATORY CONTROLLED COMPACTED FILL WITH AN ALLOWABLE BEARING CAPACITY OF 3,000 PSF. ALLOWABLE BEARING CAPACITY AT CONTINUOUS WALL FOOTINGS SHALL BE 3,000 PSF. SOIL BEARING CAPACITY SHALL BE VERIFIED BY A REGISTERED GEOTECHNICAL ENGINEER AT THE TIME OF EXCAVATION. ELEVATIONS GIVEN ARE FOR PURPOSES OF CONTRACT AND SHALL BE ADJUSTED AT THE TIME OF EXCAVATION TO MEET SOILS CONDITIONS. SITE SHALL BE PREPARED IN ACCORDANCE TO THE GEOTECHNICAL REPORT PREPARED BY "TERRACON" DATED MAY 4, 2026. PROJECT NO. 49265099
2. WALL FOOTINGS, UNLESS SHOWN OTHERWISE, SHALL BE 12" THICK WITH A 6" SPREAD EACH SIDE OF MASONRY WALL, REINFORCED WITH 2 #6 CONTINUOUS. (UNLESS NOTED OTHERWISE)
3. THE MAXIMUM WALL FOOTING STEP-OFF SHALL BE 2'-0" VERTICAL SPACED NOT LESS THAN: 4'-0" o.c. FOOTING STEP-OFFS ARE INDICATED BY THE SYMBOL: ⟋0⟍
4. WHERE PIPES OR CONDUITS RUN PERPENDICULAR TO A FOOTING, STEP TOP OF FOOTING DOWN TO ALLOW PIPES OR CONDUIT TO RUN OVER TOP OF FOOTINGS.
5. WHERE PIPES OR CONDUIT RUN PARALLEL TO A FOOTING, STEP DOWN BOTTOM OF FOOTING SO THAT A LINE DRAWN BETWEEN INVERT OF PIPE AND BOTTOM OF FOOTING SHALL NOT EXCEED 30 DEGREES, SEE "TRENCHES NEAR FOOTINGS" DETAIL.
6. NO PIPES OR CONDUIT SHALL BE PLACED IN THE FOOTINGS, OR SLAB ON GRADE.

SLAB-ON-GRADE NOTES

1. WHERE SLABS REST ON FILL, FILL SHALL BE COMPACTED IN ACCORDANCE WITH THE SPECIFICATIONS.
2. SLABS ON GROUND MAY BE PLACED IN LANE FASHION USING GALVANIZED STEEL, PREFORMED KEYED FORMS AT FLOOR JOINT LOCATION INDICATED. REINFORCING SHALL NOT CROSS CONSTRUCTION OR KEYED JOINTS.
3. SLABS-ON-GRADE REQUIRE CONTROL JOINTS PER THE TYPICAL DETAILS UNLESS NOTED OTHERWISE. LOCATE CONSTRUCTION AND CONTROL JOINTS IN SUCH A WAY TO MINIMIZE THE EFFECTS OF SHRINKAGE AND CURLING OF THE CONCRETE SLAB SECTIONS. SUBMIT TO THE ARCHITECT/ENGINEER/EQUIPMENT VENDOR THE SEQUENCE AND METHOD OF CASTING CONCRETE SLABS-ON-GRADE PRIOR TO PLACING THESE ELEMENTS.
   A. JOINT SPACING SHALL NOT EXCEED **15 FEET**, AND THE RESULTING PLAN ASPECT RATIO (PANEL LENGTH TO PANEL WIDTH) OF ANY GIVEN SECTION OF SLAB SHALL NOT EXCEED **1.5**.
   B. WHERE POSSIBLE, CONTROL JOINT LAYOUT SHALL BE SUCH THAT JOINTS ALIGN WITH THE VERTICES OF COLUMN ISOLATION BLOCKOUTS AND SHALL BE ARRANGED IN A GENERALLY ORTHOGONAL MANNER.
   C. IN AREAS WHERE NON-STRUCTURAL BONDED OVERLAY MATERIAL OR GROUT-BEDDED FLOORING ARE TO BE INSTALLED, FEATURES SUCH AS CRACK-SUPPRESSION MEMBRANES AND FLOORING DIVIDER STRIPS ALIGNED WITH SLAB CONTROL JOINTS SHALL BE DESIGNED AND INSTALLED BY OTHERS AS PART OF THE NON-STRUCTURAL FLOORING ASSEMBLY IN ORDER TO MITIGATE SUBSTRATE-INDUCED CRACKING.
4. SEE CIVIL AND ARCH. DRAWINGS AND SPECIFICATIONS FOR EXTERIOR SLAB WORK AND JOINTING. HORIZONTAL RUNS OF CONDUIT AND PIPE SHALL NOT BE PLACED IN SLABS ON GROUND, PLACE IN SUB-GRADE.
5. AT ALL TRUCK DOCK SLABS OR SERVICE COURT SLABS, PROVIDE AN 8" CONCRETE SLAB REINFORCED WITH #4 @ 12" o.c. EACH WAY, TOP AND BOTTOM. PROVIDE KEYED AND DOWELED JOINTS AT 15'-0" x 15'-0" PATTERN. TYPICAL, UNLESS NOTED OTHERWISE.
6. SLAB ON GRADE AT BUILDING SHALL BE:
   4" THICK, 3000 PSI CONCRETE REINFORCED WITH FLAT SHEETS OF WELDED WIRE FABRIC 6x6-W2.9xW2.9. SLAB SHALL BEAR ON A VAPOR BARRIER ON A NON EXPANSIVE FILL SUBGRADE AS RECOMMENDED BY THE GEOTECHNICAL ENGINEER.
7. SLAB ON GRADE AT DUMPSTER ENCLOSURE PAD SHALL BE:
   6" THICK, 4000 PSI CONCRETE REINFORCED WITH FLAT SHEETS OF WELDED WIRE FABRIC 6x6-W2.9xW2.9. SLAB SHALL BEAR ON A VAPOR BARRIER ON A NON EXPANSIVE FILL SUBGRADE AS RECOMMENDED BY THE GEOTECHNICAL ENGINEER.

COLD-FORMED STEEL NOTES

1. ALL EXTERIOR METAL STUDS ARE TO BE C.S.J. (U.N.O.)
2. ALL EXTERIOR METAL STUDS SHALL BE DESIGNED FOR LATERAL WIND PRESSURE BASED UPON ULTIMATE DESIGN WIND SPEED AS SHOWN ON DESIGN NOTES AND SHOULD BE LIMITED TO LATERAL DEFLECTION OF L/600 WHEN BACKING MASONRY, AND L/360 WHEN BACKING E.I.F.S., AND METAL PANELS
3. ALL EXTERIOR FLANGES OF STUD WALLS SHALL BE BRACED BY SHEATHING PROPERLY ATTACHED TO THEM.
4. CONTRACTOR SHALL FURNISH DETAILED DESIGN CALCULATIONS INCLUDING CONNECTION AND GAUGE OF ALL EXTERIOR METAL STUDS, METAL STUD TRUSSES AND METAL STUD TRUSS GIRDERS SEALED BY A REGISTERED ENGINEER IN THE STATE IN WHICH THE PROJECT IS BUILT.
5. THE TOP COMPRESSION FLANGES OF JOISTS & TRUSSES ARE TO BE LATERALLY BRACED BY PROPER ATTACHMENT OF THE DECKING.
6. THE TOP FLANGES OF BOTTOM CHORDS OF TRUSSES, SHALL BE BRACED LATERALLY AT INTERVALS NOT EXCEEDING 3'-0" o.c. BY CONTINUOUS C STUDS WELDED TO TOP FLANGES OR BY SHEATHING ATTACHED DIRECTLY TO BOTTOM FLANGE.
7. TOP FLANGE OF CEILING STUD SHALL BE BRACED LATERALLY BY CONTINUOUS C STUDS WELDED TO EACH JOIST. EACH BRACE SHALL BE CAPABLE OF TRANSMITTING A MINIMUM OF 200 POUNDS IN TENSION OR COMPRESSION. BOTTOM FLANGE OF CEILING SHALL BE BRACED BY SHEATHING PROPERLY ATTACHED TO BOTTOM FLANGE.
8. ALL INTERIOR FLANGES OF STUD WALLS SHALL BE BRACED EITHER BY SHEATHING OR BY CONTINUOUS CHANNELS WELDED TO FLANGE AT INTERVALS NOT EXCEEDING 2'-0" o.c. OR AS DETERMINED BY DETAIL DESIGN DOCUMENTS FURNISHED BY A REGISTERED ENGINEER IN THE STATE IN WHICH THE PROJECT IS BUILT.
9. ALL NON-LOAD-BEARING METAL STUD WALLS (INTERIOR AND EXTERIOR) SHALL BE CONNECTED TO ALLOW FOR 3/4" VERTICAL MOVEMENT BETWEEN STUD WALL AND STRUCTURE.
10. ALL OTHER INTERIOR METAL STUD PARTITIONS, BULKHEADS, CEILING STUDS AND ECT. SHALL BE FURNISHED UNDER THE DIRECTION OF THE ARCHITECTURAL DOCUMENTS . ANY ADDITIONAL STUD DESIGN REQUIREMENTS SHALL BE PROVIDED BY THE GENERAL CONTRACTOR'S METAL STUD DESIGN ENGINEER.
11. METAL STUD WALLS SHALL NOT BE ERECTED UNTIL AFTER DEAD LOADS AND ALL CONCRETE FLOORS ON FRAMING ABOVE ABOVE ARE IN PLACE.

STRUCTURAL STEEL NOTES

1. FABRICATION AND ERECTION OF ALL STEEL SHALL BE IN ACCORDANCE WITH THE AISC SPECIFICATIONS.
2. WHERE BEAMS REST ON MASONRY OR CONCRETE WALLS THEY SHALL BEAR A MINIMUM OF 8" AND SHALL HAVE BEARING PLATES AND ANCHORS.
3. WHERE NOT SHOWN OTHERWISE, ALL COLUMNS SHALL HAVE BASE PLATES AND (4) 3/4" Ø x 14" LONG ANCHOR BOLTS.
4. UNLESS SHOWN OTHERWISE, PROVIDE 5/8"x12"x7-1/2" BEARING PLATES ON 1" GROUT WITH (2) 3/4" DIAMETER ANCHOR BOLTS UNDER ALL STEEL BEAMS THAT BEAR ON MASONRY WALLS.
5. COLUMN BASES ARE DESIGNED AS UNRESTRAINED, THEREFORE COLUMNS MUST BE BRACED DURING ERECTION.
6. AS ERECTION PROGRESSES, FRAMING SHALL BE ADEQUATELY GUYED AND BRACED AND CONNECTIONS SHALL BE SECURELY BOLTED OR WELDED.
7. NO SHOP SPLICE OR OTHER CONNECTION WILL BE PERMITTED UNLESS THIS SPLICE OR DETAIL IS SHOWN ON SHOP DRAWINGS AND REVIEWED BY THE ENGINEERS.
8. ALL FASTENERS SHALL CONSIST OF ONE NUT, ONE BOLT, AND ONE WASHER.
9. ALL WELDING SHALL CONFORM TO THE "STRUCTURAL WELDING CODE - AMERICAN WELDING SOCIETY." FOR WELD TESTING REQUIREMENTS, SEE SPECIFICATIONS.
10. NO SLOTTED HOLES ARE PERMITTED UNLESS SPECIFICALLY DETAILED BY THE ENGINEER OF RECORD.
11. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO SECURE STEEL AGAINST DISPLACEMENT DURING ERECTION AND TO MAINTAIN IT AGAINST DISPLACEMENT UNTIL THE ERECTION OF ALL STEEL IS COMPLETED, ALL FLOOR AND ROOF DECKS ARE IN PLACE, AND ALL EXTERIOR MASONRY IS COMPLETED. ALL STRUCTURAL METAL WORK SHALL HAVE TEMPORARY GUYS, BRACES AND STAYS TO HOLD IT IN POSITION UNTIL IT IS PERMANENTLY SECURE.
12. WHERE ROOF DECKS FRAME ONTO A ROOF MEMBER, AND DECK IS IN A DIFFERENT PLANE THAN THE ROOF MEMBER, PROVIDE A 3/16" CONTINUOUS BENT PLATE WELDED ACROSS THE TOP OF THE ROOF MEMBER SLOPED TO MATCH PLANE OF ROOF DECK.
13. ANCHORS SHALL BE AUTOMATICALLY END WELDED WITH SUITABLE STUD WELDING EQUIPMENT IN ACCORDANCE WITH THE RECOMMENDATIONS OF THE MANUFACTURER.
14. WHERE A ROOF FRAMING MEMBER DOES NOT FRAME OVER A COLUMN, PROVIDE A 1/2" THICK CAP PLATE (i.e. TUBES AND PIPES)
15. FABRICATOR SHALL SUBMIT COMPLETE CALCULATIONS OF ALL CONNECTIONS SEALED BY AN ENGINEER LICENSED IN THE STATE IN WHICH THE PROJECT IS BUILT.
16. WHERE HSS OUTLOOKERS OCCUR PROVIDE 1/4" CAP PLATE AT END.
17. ALL EXTERIOR EXPOSED STEEL SHALL BE GALVANIZED G90 UNLESS NOTED OTHERWISE BY THE ARCHITECT.
18. ALL BEAM SHOP CONNECTIONS SHALL BE WELDED. BEAM SHOP CONNECTIONS MAY BE BOLTED IN LIEU OF WELDING PROVIDED THAT THE CONTRACTOR ASKS AND RECEIVES AN AFFIRMATIVE ANSWER BY THE ENGINEER OF RECORD IN WRITING. FURTHERMORE, ACCEPTANCE OF A BOLTED CONNECTION WILL ONLY OCCUR IF THE FABRICATOR FOLLOWS THE ECCENTRIC LOAD TABLES PROVIDED ON PAGES 7-30 THROUGH 7-53 OF THE AISC MANUAL.

CONCRETE MASONRY NOTES

1. MORTAR SHALL COMPLY WITH ASTM C 270, TYPE S, UNLESS GREATER STRENGTH IS SPECIFIED ELSEWHERE.
2. MASONRY GROUT SHALL BE COMPLYING WITH ASTM C 476, MINIMUM REQUIRED GROUT COMPRESSIVE STRENGTH f'_g = 2250 PSI, U.N.O.
3. REQUIRED MASONRY NET AREA COMPRESSIVE STRENGTH f'_m = 2250 PSI, U.N.O.
4. CONCRETE GROUT, CONFORMING TO ASTM C476, NOT MORTAR, SHALL BE USED AT CELLS AND BOND BEAMS CONTAINING REINFORCING BARS. DO NOT FILL CELLS NOT CONTAINING REINFORCING BARS, EXCEPT BELOW GRADE, UNLESS SO INDICATED.
5. UNLESS NOTED OTHERWISE, MAXIMUM CONTROL JOINT SPACING SHALL BE 32'-0", 16'-0" FROM CORNERS, COORDINATE WITH ARCHITECTURAL. ALL HORIZONTAL REINFORCING SHALL BE DISCONTINUED AT WALL CONTROL JOINTS.
6. LOAD BEARING MASONRY WALLS SHALL BE LATERALLY BRACED UNTIL ALL FLOOR / ROOF DIAPHRAGM IS IN PLACE.
7. AT ALL NON-LOAD BEARING MASONRY WALLS (INTERIOR AND EXTERIOR), PROVIDE A 3/4" CAULKED JOINT BETWEEN UNDERSIDE OF BEAM, JOIST, DECK, OR STRUCTURE AND TOP OF MASONRY WALL.
8. PROVIDE MASONRY HORIZONTAL JOINT REINFORCEMENT 16" o.c. VERTICAL IN ALL CONCRETE BLOCK WALLS. REINFORCEMENT SHALL BE FOR TOTAL WIDTH OF CAVITY WALLS.
9. PROVIDE A 2 SQUARE INCH INSPECTION HOLE AT THE BOTTOM CELL FOR EACH LIFT TO ALLOW VISUAL INSPECTION AND TO REMOVE MORTAR DROPPING PRIOR TO GROUTING.
10. ALL MASONRY SHALL BE RUNNING BOND UNLESS NOTED OTHERWISE.
11. SLEEVE ALL PLUMBING OR FIRE PROTECTION PIPING THROUGH CMU WALL.
12. SEE ARCHITECT DRAWINGS AND SPECIFICATIONS FOR MASONRY FINISHES.
13. AT LOAD-BEARING MASONRY WHERE JOISTS OR BEAMS BEAR ON MASONRY WALLS, GROUT POCKET SOLID WITH GROUT TO THE SAME FINISH FACE AS MASONRY ABOVE AND BELOW. DO THIS ONLY AFTER JOIST SEAT CONNECTION HAS BEEN INSPECTED.

POST-INSTALLED ANCHOR NOTES

THE BELOW PRODUCTS ARE THE DESIGN BASIS FOR THIS PROJECT. PRODUCT DIAMETER AND EMBEDMENT SHALL BE AS SHOWN IN THE DETAILS. INSTALL PRODUCTS IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTALLATION INSTRUCTIONS (MPII). SUBSTITUTION REQUESTS FOR PRODUCTS OTHER THAN THOSE LISTED BELOW MAY BE SUBMITTED BY THE CONTRACTOR TO THE ENGINEER-OF-RECORD (EOR) FOR REVIEW. SUBSTITUTIONS WILL ONLY BE CONSIDERED FOR PRODUCTS HAVING A CODE REPORT RECOGNIZING THE PRODUCT FOR THE APPROPRIATE APPLICATION. SUBSTITUTION REQUESTS SHALL INCLUDE CALCULATIONS THAT DEMONSTRATE THE SUBSTITUTED PRODUCT IS CAPABLE OF ACHIEVING THE EQUIVALENT PERFORMANCE VALUES OF THE DESIGN BASIS PRODUCT. CONTRACTOR SHALL CONTACT MANUFACTURER'S REPRESENTATIVE (800-999-5099) FOR PRODUCT INSTALLATION TRAINING AND A LETTER SHALL BE SUBMITTED TO THE EOR INDICATING TRAINING HAS TAKEN PLACE. SPECIAL INSPECTIONS ARE REQUIRED PER THE IBC AND ICC-ES REPORTS.

A. FOR ANCHORING INTO CONCRETE:

1. MECHANICAL ANCHORS SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ACI 355.2 AND ICC-ES AC193 FOR CRACKED CONCRETE AND SEISMIC APPLICATIONS. PRE-APPROVED ANCHORS INCLUDE:
   - SIMPSON STRONG TIE "STRONG-BOLT 2" (IAPMO-UES ESR ER-240)
2. ADHESIVE FOR REBAR AND ANCHORS SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ACI 355.4 AND ICC-ES AC308 FOR CRACKED CONCRETE AND SEISMIC APPLICATIONS. ADHESIVE ANCHORS SHALL BE INSTALLED BY A CERTIFIED ADHESIVE ANCHOR INSTALLER PER ACI 318-11 D.9.2.2 WHERE INDICATED ON THE CONTRACT DOCUMENTS. INSTALLATIONS REQUIRING CERTIFIED INSTALLERS SHALL BE INSPECTED PER ACI 318-11 D.9.2.4. PRE-APPROVED ANCHORS INCLUDE:
   - SIMPSON STRONG TIE "SET-XP" (ICC-ES ESR-2508)
   - SIMPSON STRONG TIE "AT-XP" (IAPMO-UES ESR-263)
3. POWDER-ACTUATED FASTENERS SHALL HAVE BEEN TESTED IN ACCORDANCE WITH ICC-ES AC70. PRE-APPROVED ANCHORS INCLUDE:
   - SIMPSON STRONG TIE "GAS ACTUATED PINS" (ICC-ES ESR-2811)
   - SIMPSON STRONG TIE "POWDER ACTUATED PINS" (ICC-ES ESR-2138)

B. FOR ANCHORING INTO MASONRY:

1. ANCHORAGE TO SOLID-GROUTED CONCRETE MASONRY
   a. MECHANICAL ANCHORS SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ICC ES AC01 OR ICC-ES AC106 PRE-APPROVED ANCHORS INCLUDE:
      - SIMPSON STRONG TIE "TITEN-HD" (ICC-ES ESR-1056)
      - SIMPSON STRONG TIE "STRONG-BOLT 2" (IAPMO-ES ER-240)
   b. ADHESIVE FOR REBAR AND ANCHORS SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ICC-ES AC58. PRE-APPROVED ANCHORS INCLUDE:
      - SIMPSON STRONG TIE "TITEN-HD" (ICC-ES ESR-1056)
      - SIMPSON STRONG TIE "STRONG-BOLT 2" (IAPMO-ES ER-240)
   c. POWDER-ACTUATED FASTENERS SHALL HAVE BEEN TESTED IN ACCORDANCE WITH ICC-ES AC70. PRE-APPROVED ANCHORS INCLUDE:
      - SIMPSON STRONG TIE "GAS ACTUATED PINS" (ICC-ES ESR-2811)
      - SIMPSON STRONG TIE "POWDER ACTUATED PINS" (ICC-ES ESR-2138)
2. ANCHORAGE TO HOLLOW CONCRETE MASONRY
   a. MECHANICAL ANCHORS SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ICC-ES AC106 PRE-APPROVED ANCHORS INCLUDE:
      - SIMPSON STRONG TIE "TITEN-HD"
   b. ADHESIVE FOR REBAR AND ANCHORS WITH SCREEN TUBES SHALL HAVE BEEN TESTED AND QUALIFIED FOR USE IN ACCORDANCE WITH ICC-ES AC58. THE APPROPRIATE SCREEN TUBE SHALL BE USED AS RECOMMENDED BY THE ADHESIVE MANUFACTURER. PRE-APPROVED ANCHORS WITH SCREEN TUBES INCLUDE:
      - SIMPSON STRONG TIE "SET" (ICC-ES ESR-1772)
   c. POWDER-ACTUATED FASTENERS SHALL HAVE BEEN TESTED IN ACCORDANCE WITH ICC-ES AC70. PRE-APPROVED ANCHORS INCLUDE:
      - SIMPSON STRONG TIE "GAS ACTUATED PINS" (ICC-ES ESR-2811)
      - SIMPSON STRONG TIE "POWDER ACTUATED PINS" (ICC-ES ESR-2138)

C. FOR ANCHORING INTO STEEL POWDER-ACTUATED FASTENERS SHALL HAVE BEEN TESTED IN ACCORDANCE WITH ICC-ES AC70. PRE-APPROVED ANCHORS INCLUDE:

1. SIMPSON STRONG TIE "GAS ACTUATED PINS" (ICC-ES ESR-2811)
2. SIMPSON STRONG TIE "POWDER-ACTUATED PINS" (ICC-ES ESR-2138)

STEEL BAR JOIST NOTES

1. JOIST BRIDGING SHALL NOT BE USED TO SUPPORT CONDUIT, PIPING, DUCTWORK, ETC.
2. JOISTS SHALL NOT BE FIELD MODIFIED EXCEPT AS SHOWN.
3. WHERE JOISTS CANNOT ACHIEVE THE REQUIRED BEARING ON STEEL BEAMS, STAGGER JOISTS TO PROVIDE ADEQUATE BEARING.
4. STEEL JOISTS SHALL BE OPEN WEB STEEL JOISTS OF THE SIZES AND SERIES SHOWN ON THE DRAWINGS. JOISTS, BRIDGING AND SPACING OF BRIDGING SHALL MEET THE REQUIREMENTS OF THE LATEST EDITION OF THE "STANDARD SPECIFICATIONS" OF THE STEEL JOIST INSTITUTE, EXCEPT WHERE OTHERWISE INDICATED BY THE DRAWINGS OR SPECIFICATIONS.
5. WHERE ANGLE BRACES ARE SHOWN ON STRUCTURAL SECTIONS, JOIST MANUFACTURER SHALL RESOLVE AN AXIAL LOAD OF 2000 POUNDS FROM THE BRACE INTO THE JOIST - TYPICAL UNLESS NOTED OTHERWISE.
6. IN ADDITION TO WHAT IS CALLED FOR ON PLAN, BAR JOISTS SHALL BE DESIGNED TO SUPPORT AN ADDITIONAL CONCENTRATED LOAD OF 300 POUNDS AT TOP OR BOTTOM CHORD AT ANY ONE LOCATION ALONG THE SPAN.
7. AT THE END OF EACH ROOF JOIST, PROVIDE A CONTINUOUS ROW OF BRIDGING AT THE LAST BOTTOM CHORD PANEL POINT FOR UPLIFT. TYPICAL AT EACH END OF JOIST.
8. NET WIND UPLIFT SHALL BE CONSIDERED USING THE ASD COMPONENTS AND CLADDING PRESSURES (THIS SHEET) AND A DEAD LOAD OF 8 PSF.

REPAIR, PROTECT, AND STRENGTHENING NOTES

THE BELOW PRODUCTS ARE THE DESIGN BASIS FOR THIS PROJECT. CONTRACTOR SHALL FOLLOW MANUFACTURER'S INSTALLATION INSTRUCTIONS AND CONTACT MANUFACTURER'S REPRESENTATIVE (800-999-5099) WITH PRODUCT RELATED QUESTIONS. SUBSTITUTION REQUESTS FOR PRODUCTS OTHER THAN THOSE LISTED BELOW MAY BE SUBMITTED BY THE CONTRACTOR TO THE ENGINEER-OF-RECORD (EOR) FOR REVIEW AND APPROVAL. SUBSTITUTIONS WILL ONLY BE CONSIDERED FOR PRODUCTS HAVING INDEPENDENT TEST RESULTS OR OTHER DOCUMENTATION INDICATING THE PRODUCT IS APPROPRIATE FOR THE INTENDED APPLICATION.

A. REBAR PRIMER: REBAR PRIMER SHALL BE USED TO PROTECT EXISTING STEEL REINFORCING AND ENCOURAGE POSITIVE BOND FROM EXISTING STEEL REINFORCING TO NEW REPAIR MATERIAL. PRE-APPROVED PRODUCTS INCLUDE:
   1. SIMPSON STRONG TIE "FX-406"

B. BONDING AGENTS: BONDING AGENTS SHALL BE USED TO ENCOURAGE POSITIVE BOND OF NEW REPAIR MATERIAL TO EXISTING CONCRETE. PRE-APPROVED PRODUCTS INCLUDE:
   1. SIMPSON STRONG TIE "FX-762"

C. REPAIR MORTARS: REPAIR MATERIAL SHALL BE USED TO REPAIR AREAS OF DAMAGED CONCRETE. PRE-APPROVED PRODUCTS INCLUDE:
   1. SIMPSON STRONG TIE "FX-263" (FOR USE IN OVERHEAD AND VERTICAL APPLICATIONS)
   2. SIMPSON STRONG TIE "FX-261" (FOR USE IN HORIZONTAL AND FORM & POUR APPLICATIONS)

D. CRACK REPAIR SYSTEM: CRACK REPAIR SYSTEM SHALL CONSIST OF CRACK INJECTION MATERIAL AND PASTE OVER ADHESIVE AND SHALL BE USED TO PRESSURE INJECT CRACKS. PRE-APPROVED SYSTEMS INCLUDE:
   1. SIMPSON STRONG TIE "FX-751 LV" INJECTION MATERIAL WITH "FX-763" PASTE OVER ADHESIVE

E. NON-SHRINK GROUT MATERIAL: NON-SHRINK GROUT MATERIAL SHALL BE USED TO GROUT BENEATH BASEPLATES, BEARING PLATES AND EQUIPMENT BASES. PRE-APPROVED PRODUCTS INCLUDE:
   1. SIMPSON STRONG TIE "FX-228"

SUBMITTAL NOTES

1. THE GENERAL CONTRACTOR SHALL SUBMIT A SCHEDULE OF SUBMITTALS PRIOR TO CONSTRUCTION BEGINNING ON THE PROJECT. THE SCHEDULE SHOULD DESCRIBE WHAT EACH SUBMITTAL IS, WHETHER IT IS THE ENTIRE PACKAGE, OR BROKEN INTO PHASES FOR REVIEW.
2. THE GENERAL CONTRACTOR SHALL ALLOW FOR 10 BUSINESS DAYS OF ALL STRUCTURAL SUBMITTALS. IF THE CONTRACTOR WISHES TO EXPEDITE OR REDUCE THE REVIEW TIME, IT SHALL BE DONE FOR AN ADDITIONAL CHARGE AND MUST BE NEGOTIATED WITH THE ARCHITECT AND ENGINEER PRIOR TO THE SUBMISSION OF THE PACKAGE.
3. IN THE EVENT THE CONTRACTOR ATTEMPTS TO SUBMIT THE ENTIRE PROJECT AT THE SAME TIME FOR SUBMISSION, ADDITIONAL TIME WILL BE REQUIRED BEYOND THE STANDARD 10 DAY REVIEW TIME.
4. WHERE GLASS REQUIRES ADDITIONAL DESIGN OF MISCELLANEOUS STEEL AND MULLIONS, CONTRACTOR SHALL FURNISH DETAILED DESIGN CALCULATIONS SEALED BY A REGISTERED ENGINEER IN THE STATE IN WHICH THE PROJECT IS BUILT.
5. STRUCTURAL STEEL SUPPLIER SHALL PROVIDE COMPLETE SHOP DRAWINGS INCLUDING AN ANCHOR SETTING PLAN, ERECTION DRAWINGS WITH ALL PIECE MARKS, AND ASSOCIATED FABRICATION DRAWINGS SHOWING ALL MATERIAL AND CONNECTIONS. CONNECTIONS SHALL BE DESIGNED BASED UPON THE END SHEAR TABLES PROVIDED IN THE AISC UNLESS NOTED OTHERWISE.
6. REBAR SUPPLIER SHALL PROVIDE COMPLETE SHOP DRAWINGS INCLUDING PLAN VIEWS WITH MARKS ILLUSTRATING BAR SIZES AND BAR BENDING DETAILS. INCLUDE ALL LOCATION OF STEPS NOTED IN THE GENERAL NOTES FOR PLUMBING AND CIVIL LINES WHERE APPLICABLE. WHERE CONCRETE PIER AND STEM WALLS OCCUR PROVIDE SECTION CUTS, AND ELEVATIONS ILLUSTRATING THE MATERIAL SPECIFIED
7. REBAR SUPPLIER SHALL PROVIDE COMPLETE SHOP DRAWINGS FOR LOAD BEARING MASONRY. PROVIDE ALL WALLS INCLUDING NON LOAD BEARING PARTITIONS IN ELEVATIONS DEPICTING THE VERTICAL WALL REINFORCEMENT/ LAPS, ALL LINTELS AT OPENINGS , REINFORCING AND BOND BEAMS AS DESCRIBED IN TYPICAL DETAILS AND WALL SECTIONS INCLUDING MASONRY CONTROL JOINT LOCATIONS.
8. JOIST AND DECK SUPPLIER SHALL PROVIDE COMPLETE SHOP DRAWINGS INCLUDING ERECTION DRAWINGS, SECTIONS INCLUDING MATERIALS PROVIDED, ALL ASSOCIATED CONNECTIONS AS OUTLINED IN THE CONTRACT DOCUMENTS AND SPECIFICATIONS. SHOP DRAWINGS SHALL INCLUDE ALL EDGE OF DECK DIMENSIONS, EDGE OF JOIST DIMENSIONS .
9. SEE DISCIPLINE SPECIFIC NOTES AND SPECIFICATIONS FOR ALL ADDITIONAL SUBMITTAL REQUIREMENTS.
10. FOR THIRD-PARTY DESIGN SUBMITTALS INCLUDING, BUT NOT LIMITED TO, PRECAST, HOLLOW CORE SLABS, METAL STUDS, STAIRS & HANDRAILS, GLASS DESIGN, AND WOOD TRUSSES, PLEASE SUBMIT MANUFACTURER'S DRAWINGS WITH ACCOMPANYING CALCULATIONS SEALED BY A REGISTERED ENGINEER IN THE STATE IN WHICH THE PROJECT IS BUILT.

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Rev	Date	Comments
1	07/02/2026	PERMIT SET

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